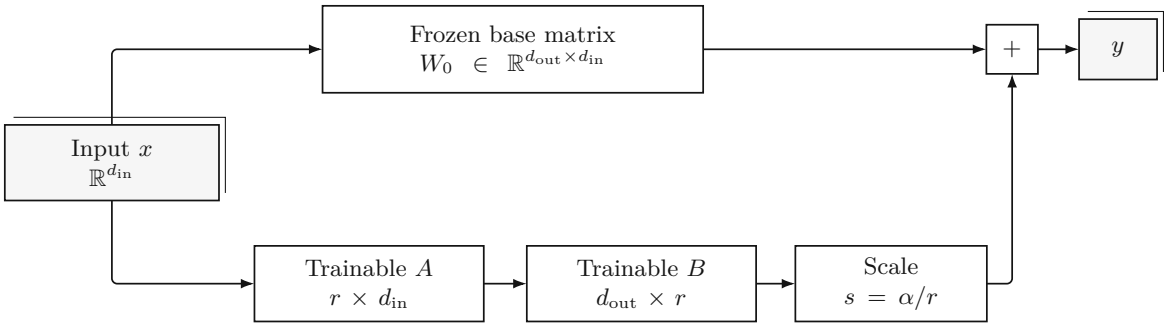
LoRA: a frozen dense path plus a trainable low-rank update

Conceptual schematic; symbolic dimensions; batch axis omitted



$$y = W_0 x + sBAx; \Delta W = sBA \text{ and } r \ll \min(d_{\text{in}}, d_{\text{out}}).$$

The adapter branch has a rank bottleneck; its scaled output adds to the frozen base-layer output.